

Optimization of Water Distribution Networks Using Genetic Algorithms

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1 State of the Art

The optimization of water distribution networks (WDNs) is an important research topic because the construction and maintenance of these systems involve high costs. A water distribution network consists of pipes, reservoirs, pumps, and valves that transport water from sources to consumers. The goal of optimization is usually to minimize the total cost of the network while ensuring that hydraulic constraints such as pressure and flow requirements are satisfied.

Because the problem involves many possible pipe configurations and nonlinear hydraulic constraints, it is considered a complex combinatorial optimization problem. Over time, several approaches have been proposed to solve it.

1.1 Mathematical Programming Approaches

In mathematical programming approaches, the network design problem can be formulated as an optimization model.

The objective is to minimize the total cost of the pipes in the network:

$$\min C = \sum_{i=1}^N c_i L_i$$

where:

- C – total network cost
- c_i – cost per unit length of pipe i
- L_i – length of pipe i
- N – total number of pipes

The network must also satisfy hydraulic constraints, such as flow conservation at each node:

$$\sum Q_{in} = \sum Q_{out}$$

where Q represents the water flow entering or leaving a node.

In particular, the conservation of mass at each node and the conservation of energy along each pipe are expressed through nonlinear equations. Integer variables are used to represent the selection of pipe diameters from a set of commercially available options. The objective function typically minimizes the total installation cost of the network while ensuring that all pressure and demand requirements are satisfied.

For example, the integer linear programming approach proposed by Samani [4] models pipe diameter selection as a discrete decision problem and uses optimization techniques to determine the least-cost network configuration. Although these methods can provide exact solutions for smaller networks, their computational complexity increases rapidly as the network size grows.

1.2 Genetic Algorithms

Genetic algorithms are inspired by the process of natural evolution. They work by generating a population of possible solutions and improving them through operations such as selection, crossover, and mutation. Each solution represents a possible configuration of pipe diameters or network parameters.

All candidate solutions are evaluated using a fitness function. The fitness value is usually based on the cost of the network:

$$Fitness = \frac{1}{C + P}$$

where:

- C – total network cost
- P – penalty for violating hydraulic constraints

Solutions with lower cost and fewer constraint violations receive higher fitness values and are more likely to be selected for the next generation.

Studies have shown that genetic algorithms are effective in exploring large search spaces and finding near-optimal solutions for complex network design problems [1].

1.3 Hybrid Optimization Methods

Some studies propose hybrid approaches that combine genetic algorithms with mathematical programming techniques. For example, a genetic algorithm can explore the search space while integer linear programming helps refine the solutions [3].

These hybrid approaches attempt to combine the global search capabilities of evolutionary algorithms with the precision of mathematical optimization methods.

Overall, the literature shows that metaheuristic algorithms, especially genetic algorithms, are among the most effective approaches for optimizing water distribution networks.

1.4 Reinforcement Learning Approaches

Reinforcement learning (RL) is a machine learning approach where an agent learns how to control a system by interacting with it and receiving rewards based on its actions. Recent research shows that deep reinforcement learning (DRL) can be useful for managing complex urban water systems such as water distribution networks and stormwater systems [6].

In this approach:

- The **agent** observes the current **state** of the system (e.g., pressure, flow, or tank levels).
- The agent selects an **action** using a neural network policy.
- The action is applied to the **environment** (the water network).
- The environment returns a **reward** and a **new state**.
- The agent uses this information to improve its decision-making policy over time.

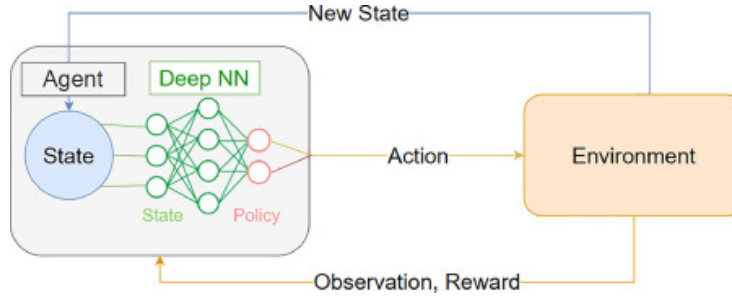


Figure 1: [6]

By repeatedly interacting with the network and receiving rewards, the algorithm learns a control policy that improves the efficiency and reliability of the water system.

The reward function is typically designed to encourage desirable system behaviour, such as minimizing operational cost and energy consumption while maintaining acceptable pressure levels in the network. For example:

$$R = -C - \lambda P_{violations}$$

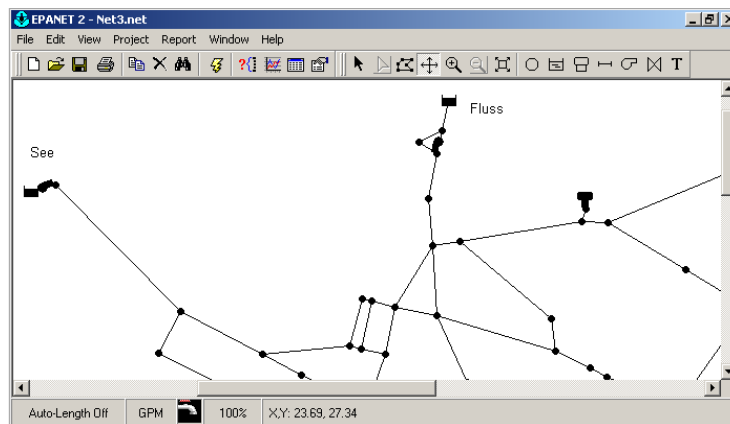
where C represents the operational cost and $P_{violations}$ represents penalties for violating hydraulic constraints. Recent studies show that deep reinforcement learning algorithms can effectively learn control strategies for complex water distribution systems.

1.5 Algorithms Combined with Hydraulic Simulation

Another common approach is combining optimization algorithms with hydraulic simulation tools such as EPANET. In this case, the optimization algorithm generates candidate solutions, and the simulation software evaluates whether they satisfy hydraulic constraints.

This integration allows researchers to test many possible network designs while ensuring that pressures and flows remain within acceptable limits. Such hybrid methods are widely used in modern research because they allow realistic modelling of water distribution systems.

Various evolutionary and metaheuristic algorithms have been applied in this framework, including Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Simulated Annealing (SA), and Harmony Search (HS). These algorithms explore large solution spaces efficiently and are well suited for the discrete and nonlinear nature of water distribution network design problems.



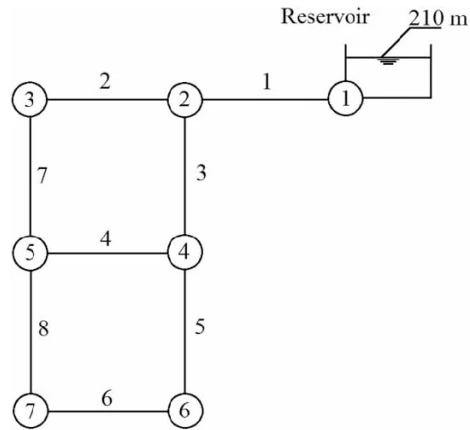
2 Benchmark Instances

To evaluate and compare optimization algorithms, researchers use standard benchmark networks. These benchmark instances represent simplified but realistic water distribution systems and allow different algorithms to be tested under the same conditions.

2.1 Two-Loop Network

The Two-Loop network is one of the most commonly used benchmark problems. It is a small network composed of two loops and several pipes with different possible diameters.

Although the network is relatively simple, it is useful for testing optimization algorithms and evaluating how well they handle hydraulic constraints.



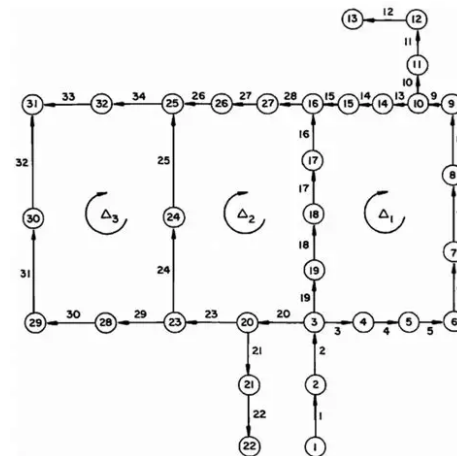
Quirks:

- Very small search space
- Looped structure with multiple flow paths
- Discrete pipe diameter choices

2.2 Hanoi Network

The Hanoi network is another well-known benchmark used in many studies. It represents a planned trunk water distribution system in Hanoi, Vietnam.

This network contains 32 nodes, 34 pipes, and one reservoir. Because of its moderate size and looped structure, it is frequently used to evaluate genetic algorithms and other metaheuristic methods [1].



Quirks:

- Much larger search space than simple benchmarks
- Multiple loops affecting pressure distribution
- Strict minimum pressure requirements at nodes

2.3 New York City Tunnel Network

The New York City tunnel network is a larger and more complex benchmark problem. It represents the design of additional water tunnels in the New York City water supply system.

The objective is to minimize construction costs while ensuring that the pressure requirements are satisfied throughout the network.

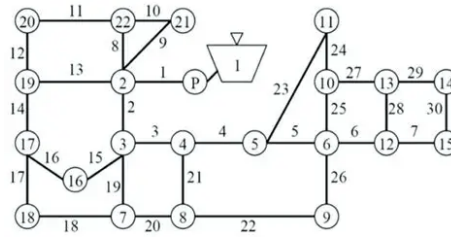


Quirks:

- Focus on expanding an existing network
- High construction cost trade-offs
- Strategic tunnel placement affects whole system

2.3.1 GoYang Network

The GoYang network represents a medium-sized water distribution system that has been used in several optimization studies. It contains multiple loops and a larger number of pipes compared to smaller benchmarks such as the Two-Loop network. Because of its increased complexity, the GoYang network is often used to test the scalability and efficiency of optimization algorithms.



Quirks:

- Medium-size network with complex topology
- Many interacting loops
- Higher computational complexity

2.3.2 Balerna Network

The Balerna network is one of the largest benchmark systems used in water distribution optimization studies. It represents a real irrigation network located in Spain and contains hundreds of pipes and nodes. Due to its large size and complexity, the Balerna network is often used to evaluate the performance of optimization algorithms on large-scale problems.



Figure 2: Balerna Network

Quirks:

- Very large real-world irrigation network
- Hundreds of pipes and nodes
- Used to test scalability of optimization algorithms

Benchmark networks are often provided as EPANET input files containing information about pipes, nodes, demands, and hydraulic parameters. Using standardized benchmarks allows researchers to compare different optimization methods in a consistent and reproducible way.

References

- [1] M. Creaco, G. Pezzinga. *Multiobjective Optimization of Water Distribution Networks Using Genetic Algorithms*. Water, 14(6), 2022.
Available at: <https://www.mdpi.com/2073-4441/14/6/851>
- [2] A. Martínez et al. *Optimization of Water Distribution Networks Using Evolutionary Algorithms*.
Available at: <https://rua.ua.es/server/api/core/bitstreams/92c6f0dc-5320-407a-a78c-f4785cb4bf39/content>
- [3] A. Afshar, B. Marino. *GA-ILP Method for Optimization of Water Distribution Networks*.
Available at: <https://d1wqtxts1xzle7.cloudfront.net/67387871/s11269-011-9775-420210520-3046-1w7t3fv-libre.pdf>
- [4] H. Mohammad Vali Samani. *Optimization of Water Distribution Networks Using Integer Linear Programming*.
Available at: https://www.researchgate.net/profile/Hosseini-Mohammad-Vali-Samani/publication/245297377_Optimization_of_Water_Distribution_Networks_Using_Integer_Linear_Programming/links/558c172b08ae40781c2036b5/Optimization-of-Water-Distribution-Networks-Using-Integer-Linear-Programming.pdf
- [5] I. Sârbu. *Water Distribution Systems – Analysis and Optimization*.
Available at: <https://academiaromana.ro/sectii2002/proceedings/doc2010-4/06-Sarbu.pdf>
- [6] A. Negm, X. Ma, G. Aggidis. *Deep reinforcement learning challenges and opportunities for urban water systems*. Water Research, Volume 253, 2024, Article 121145.
Available at: <https://www.sciencedirect.com/science/article/pii/S0043135424000459>